1 Convolution

1.1 Causal convolution using quantics MPS

We consider the causal convolution

$$F(t) = \int_{T_0}^{t} K(t-v)S(v)\,dv, \qquad T_0 \le t < T_M, \tag{1}$$

where K is the convolution kernel and S is the signal.

Discretization. Let $L = 2^N$ be the number of grid points, and define

$$\Delta t = \frac{T_M - T_0}{L}. \tag{2}$$

The signal and the output are discretized on the same grid:

$$t_m = T_0 + m\Delta t, \qquad m = 0, \dots, L-1. \tag{3}$$

The kernel is defined on the corresponding time-lag grid:

$$\tau_k = k\Delta t, \qquad k = 0, \dots, L-1. \tag{4}$$

Thus, the discrete functions are

$$S_m = S(t_m), \qquad K_k = K(\tau_k), \qquad F_m = F(t_m). \tag{5}$$

Scaling the inclusive convolution gives the following uncorrected quadrature expression; its endpoint convention is discussed below:

$$\boxed{F_m \approx \Delta t \sum_{j=0}^{m} K_{m-j} S_j.} \tag{6}$$

Endpoint convention and quadrature corrections. The current `conv` operation returns the inclusive discrete sum

$$C_m = \sum_{j=0}^{m} K_{m-j} S_j, \tag{7}$$

without the factor Δt. This is a discrete convolution, not yet a particular continuous-integration rule. The interval $[T_0, t_m]$ contains m grid intervals but $m+1$ samples. For example, if $K = S = 1$, the integral is $m\Delta t$, whereas the uncorrected inclusive sum gives $(m+1)\Delta t$. In particular, the uncorrected value at $m = 0$ need not vanish.

To reproduce the previous exclusive-prefix convention in the time-lag coordinate, use the right-rectangle rule in the physical signal time v:

$$\boxed{F_m^{\mathrm{R}} = \Delta t \sum_{j=1}^{m} K_{m-j} S_j = \Delta t \left(C_m - S_0 K_m\right).} \tag{8}$$

Indeed, writing $k = m - j$ gives an exclusive sum over lag indices $k = 0, \dots, m-1$. Consequently:

- For an adiabatic activation $S(v) = \lambda(v)$ with $S_0 = \lambda(T_0) = 0$, the correction vanishes: $F_m^{\mathrm{R}} = \Delta t\, C_m$.
- For the instantaneous-quench convention $S_j = 1$, including $S_0 = 1$, one obtains $F_m^{\mathrm{R}} = \Delta t(C_m - K_m) = \Delta t \sum_{k=0}^{m-1} K_k$, exactly the previous exclusive prefix sum.

The corrected result satisfies $F_0^{\mathrm{R}} = 0$ in both cases.

Alternatively, trapezoidal quadrature assigns half weight to both endpoints:

$$\boxed{F_m^{\mathrm{trap}} = \Delta t \left[C_m - \frac{K_0 S_m + S_0 K_m}{2}\right].} \tag{9}$$

This expression also gives $F_0^{\mathrm{trap}} = 0$. The endpoint corrections require only scalar multiplications and additions of MPSs; they do not require changing the convolution algorithm. The factor Δt is applied once to the output, not separately to each input.

Convolution using the addition tensor. Following Section 10.3 of *Who can compete with quantum computers?*, we introduce the addition tensor

$$\Theta_{mjk} = \delta_{m,j+k}, \tag{10}$$

where all indices belong to $\{0, \ldots, L-1\}$.

The discrete convolution can then be expressed as

$$\boxed{F_m \approx \Delta t \sum_{j,k=0}^{L-1} \Theta_{mjk} S_j K_k.} \tag{11}$$

The Kronecker delta enforces $m = j + k$, so that only terms satisfying $j \leq m$ contribute.

The addition tensor admits a low-rank matrix product operator representation. By contracting this tensor with the two quantics MPS representing S and K, we obtain the MPS representation of F.

Output grid and boundary conditions. Since S, K, and F each contain $L = 2^N$ samples, all three can be represented using N binary indices.

The physical grids satisfy

$$\boxed{\begin{aligned} \mathcal{G}_S &= [T_0, T_M[, \\ \mathcal{G}_K &= [0, T_M - T_0[, \\ \mathcal{G}_F &= [T_0, T_M[. \end{aligned}} \tag{12}$$

All three grids have the same spacing Δt.

The addition MPO must impose a zero final carry, thereby excluding overflow and preventing periodic wraparound.

Consequently, no zero-padding is required to calculate the causal convolution on the original output grid.

The tensor-network contraction implements the discrete convolution exactly in the absence of MPS truncation. The remaining numerical errors arise from the quadrature approximation and possible MPS compression.

Special case $T_0 = 0$. When $T_0 = 0$, the three grids coincide:

$$\boxed{\mathcal{G}_S = \mathcal{G}_K = \mathcal{G}_F = [0, T_M[.} \tag{13}$$

1.2 Evaluating the integrated response

The convolution output already represents the integral up to the observation time:

$$F(t) = \int_{T_0}^{t} K(t-v) S(v)\, dv. \tag{14}$$

Once the convolution, grid-spacing factor, and chosen endpoint treatment have been applied, evaluating this output MPS at t gives the desired integral. No second integration is needed. With $F(T_0) = 0$, subtracting $F(T_0)$ is unnecessary; it simply gives $F(t) - F(T_0) = F(t)$. The evaluation time must belong to the represented grid: the upper boundary T_M is excluded and requires separate endpoint handling if needed.

This output is not, in general, a primitive for integration over arbitrary subintervals with a fixed observation time. Such a primitive would be

$$P_t(x) = \int_{T_0}^{x} K(t-v) S(v)\, dv, \tag{15}$$

with t held fixed while x varies. Although $P_t(t) = F(t)$, generally $P_t(x) \neq F(x)$, since the kernel in $F(x)$ is $K(x-v)$. Consequently,

$$F(b) - F(a) \neq \int_{a}^{b} K(b-v) S(v)\, dv \tag{16}$$

in general. This does not prevent using the convolution output directly for the intended integral from the initial time to the observation time.

1.3 Application to the first-order Green function

For the relative-time convention $v = t - t_1 \geq 0$, the required contribution is

$$G^{(1)}(t = t') = iU \int_0^{t-T_0} \lambda(t - v)\widetilde{G}^{(1)}(0, 0; -v)\, dv. \tag{17}$$

To distinguish the roles of the two inputs from the earlier notation, write the saved integrand as the signal on the positive-delay grid:

$$S_{\mathrm{GF}}(v) = i\widetilde{G}^{(1)}(0, 0; -v). \tag{18}$$

The saved `t_ti` integrand MPS already contains both the negative internal-time argument and the prefactor i. Neither an additional reflection nor another multiplication by i is required.

Introduce elapsed observation time $T = t - T_0$ and the activation input

$$A(w) = \lambda(T_0 + w), \qquad w \geq 0. \tag{19}$$

Then the desired response is a convolution of these two inputs:

$$\boxed{R(T) = \int_0^T S_{\mathrm{GF}}(v)A(T - v)\, dv, \qquad G^{(1)}(t = t') = UR(t - T_0).} \tag{20}$$

Both input arrays are indexed by elapsed coordinates on $[0, T_M - T_0[$, with the same number of points and spacing Δt. Thus, after applying the quadrature convention, a helper such as `make_indefinite_integral_conv` is intended to construct

$$R_m \approx \Delta t \sum_{j=0}^{m} S_{\mathrm{GF}}(j\Delta t)A((m - j)\Delta t). \tag{21}$$

For an adiabatic ramp with $A(0) = \lambda(T_0) = 0$, this inclusive sum already equals the exclusive rectangle sum over lag indices $j < m$; its diagonal endpoint term vanishes and $R_0 = 0$.

The final calculation therefore consists only of evaluating the response MPS and multiplying by U. If its returned grid represents elapsed time, evaluate it at $t - T_0$. If the same array is assigned the physical-time grid $[T_0, T_M[$ instead, evaluate it at t. For $T_0 = 0$, these coordinates coincide. The grid-spacing factor is applied during response construction, not again during evaluation, and there is no need to subtract the origin value for the ramp with $\lambda(T_0) = 0$.